

- (a) A packet contains a large number of flower seeds that produce white, yellow or red flowers. Half of the seeds produce white flowers and one third produce yellow flowers. The remainder of the seeds produce red flowers.

(i) Two seeds are chosen at random. Find the probability that

(a) one will produce a yellow flower and one will produce a white flower, [2]

(b) neither will produce a red flower. [2]

(ii) Three seeds are chosen at random. Find the probability that at most two seeds will produce white flowers. [2]

(iii) Given that n seeds are chosen at random, find the probability that all the seeds will produce red flowers. [1]

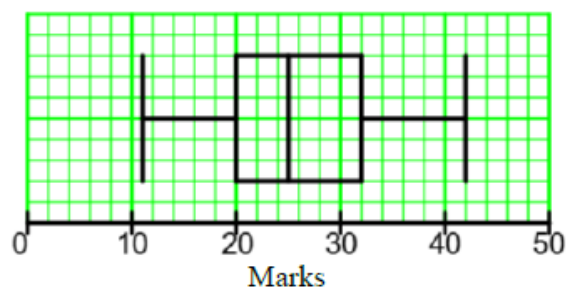
- (b) The stem and leaf diagram below shows the marks attained by 23 students in Class X for a Science quiz.

1	5	6	8	9	9	9		
2	0	0	1	1	2	3	4	7
3	3	4	4	5	6	8		
4	0	1	2					

Key : 1 | 5 means 15 marks

(i) Using the data given, find the median mark. [1]

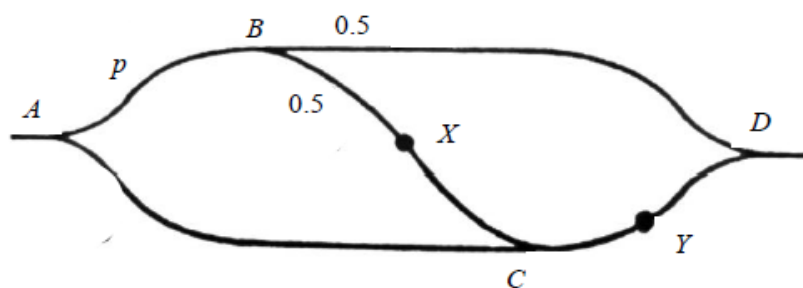
(ii) The box-and-whisker plot shows the distribution of the results of students in Class Y for the same Science quiz.



Find the interquartile range. [1]

(iii) Make two comparisons of the performance of the two classes X and Y for the Science quiz. [2]

2	SST/2025/I/10
	A bag contains some blue marbles, some green marbles and some red marbles. The probability of choosing a green marble is 0.2 . The probability of choosing a red marble is 0.35 . (a) Find the probability of choosing a blue marble at random. [1] (b) In the bag, there are 9 more red marbles than green marbles. Find the total number of marbles in the bag. [2]
3	SST/2025/II/07
	(b) Three boxes, X, Y and Z contain the following balls. Box X: 2 black balls and 1 white ball. Box Y: 1 black ball and 5 white balls. Box Z: 2 black balls and 2 white balls. A ball is picked at random from a box. The box is chosen by throwing a die and applying the following rules. 1. If the outcome on the die is 6, box X will be chosen. 2. If the outcome on the die is a 4 or a 5, box Y will be chosen. 3. If the outcome on the die is a 1, a 2 or a 3, box Z will be chosen. If a black ball is obtained, a prize of \$10 is awarded. There will be a \$2 fee to play the game. (i) Complete the tree diagram. (ii) Find the probability of winning the \$10 prize. [3] (iii) Find the amount gained or lost after playing 12 games. [2]



A road system has two junctions, A and D , where the traffic is moving from A to D . At A , the probability that a car travels along the road AB is p . At B , the probability that a car travels along the roads BD or BC is 0.5 in each case.

Find, in terms of p , the probability that,

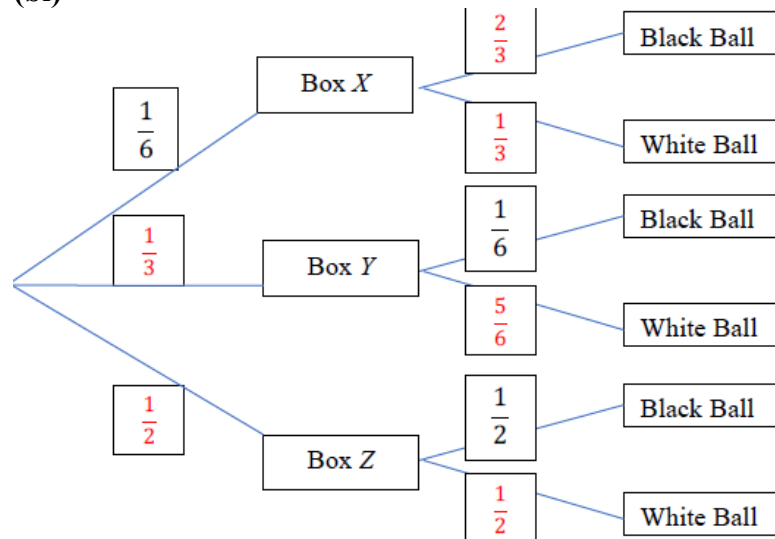
- (a) the car travels along AC . [1]
- (b) a car travelling from A to D
 - (i) will pass the point X , [1]
 - (ii) will pass the point Y . [2]

Solution

1	(ai)(a) $P(1 \text{ yellow}, 1 \text{ white}) = \left(\frac{1}{2} \times \frac{1}{3}\right) + \left(\frac{1}{3} \times \frac{1}{2}\right)$ $= \frac{1}{3}$ (ai)(b) $P(\text{neither red}) = \frac{5}{6} \times \frac{5}{6}$ $= \frac{25}{36}$ (aii) $P(\text{at most 2}) = 1 - P(\text{all white})$ $= 1 - \left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right)$ $= \frac{7}{8}$ (aiii) $P(\text{all red}) = \left(\frac{1}{6}\right)^n$ (bi) 23 (bii) 32 – 20 = 12 (biii) 1. Class Y's students scored better on average, as Class Y has a higher median (25) than that of Class X (23). 2. The marks of Class Y's students are more consistent than that of Class X's, as Class Y (12) has a lower interquartile range than Class X (35 – 19 = 16).
2	(a) $P(\text{blue marble})$ $= 1 - 0.35 - 0.2$ $= 0.45$ 45% : not accepted (b) $\text{Total number of marbles in the bag}$ $= \frac{9}{0.35 - 0.2}$ $= 60$

3

(bi)



(bii)

$$\frac{1}{6} \times \frac{2}{3} + \frac{1}{3} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} = \frac{5}{12}$$

(biii)

5 wins in 12 games gains \$50
 Fee incurred = \$2 (12) = \$24
 Amount earned = \$50 - \$24
 = \$26 gained

4

(a)

$$1 - p$$

(bi)

$$0.5p$$

(bii)

$$(1 - p) + 0.5p = 1 - 0.5p$$